ClipSAT : ACT International Subject Test: Mathematics Level 2 [ANSWER KEY]

Unit 5.01: Trigonometry, Identities & Polar Coordinates

Q1. A point has polar coordinates r, theta = left6, frac{2pi}{3}right. What are the Cartesian coordinates x, y of this point?

[x] A: -3, 3sqrt{3}

[ ] B: -3sqrt{3}, 3

[ ] C: 3, -3sqrt{3}

[ ] D: 3sqrt{3}, -3

Solution: To convert polar coordinates r, theta to Cartesian coordinates x, y, use the conversion formulas: x = r costheta quad text{and} quad y = r sintheta Given r = 6 and theta = frac{2pi}{3}: x = 6 cosleftfrac{2pi}{3}right = 6 left-frac{1}{2}right = -3 y = 6 sinleftfrac{2pi}{3}right = 6 leftfrac{sqrt{3}}{2}right = 3sqrt{3} Thus, the Cartesian coordinates are -3, 3sqrt{3}. Common errors: - Interchanging x and y using x = rsintheta and y = rcostheta yields 3sqrt{3}, -3. - Reversing the quadrant signs or applying frac{pi}{6} reference angle formulas erroneously produces -3sqrt{3}, 3.

Q2. In triangle ABC, the length of side AB is x, the length of side BC is x + 3, and the length of side AC is 7. If angle B = 60^circ, what is the value of x?

[x] A: 5

[ ] B: 8

[ ] C: 4

[ ] D: 6

Solution: To find x, apply the Law of Cosines to triangle ABC with angle B: AC^2 = AB^2 + BC^2 - 2ABBCcosB Substitute the given values AC = 7, AB = x, BC = x + 3, and angle B = 60^circ: 7^2 = x^2 + x + 3^2 - 2xx + 3cos60^circ Recall that cos60^circ = frac{1}{2}: 49 = x^2 + x^2 + 6x + 9 - 2xx + 3leftfrac{1}{2}right 49 = 2x^2 + 6x + 9 - xx + 3 49 = 2x^2 + 6x + 9 - x^2 - 3x 49 = x^2 + 3x + 9 Rearrange into standard quadratic form: x^2 + 3x - 40 = 0 Factor the quadratic expression: x - 5x + 8 = 0 This yields x = 5 or x = -8. Since a triangle side length must be positive, we reject x = -8, giving x = 5. Common traps: - Solving for the length of side BC instead of x gives x + 3 = 8 Choice B. - Sign errors when expanding x - 5x + 8 = 0 often lead students to select x = 8.

Q3. In the polar coordinate plane, points A and B lie on the curve r = 4sin2theta such that their polar angles satisfy 0 < theta_A < theta_B < frac{pi}{2} and theta_B - theta_A = frac{pi}{3} . What is the maximum possible area of triangle OAB , where O is the origin?

[x] A: sqrt{3}

[ ] B: 2

[ ] C: 2sqrt{3}

[ ] D: 4sqrt{3}

Solution: In polar coordinates, the area of a triangle formed by the pole O and two points Ar_A, theta_A and Br_B, theta_B is given by: text{Area} = frac{1}{2} r_A r_B sintheta_B - theta_A Given theta_B - theta_A = frac{pi}{3} , we have: text{Area} = frac{1}{2} r_A r_B sinleftfrac{pi}{3}right = frac{sqrt{3}}{4} r_A r_B For points on the curve r = 4sin2theta , since 0 < theta_A < theta_B < frac{pi}{2} , both 2theta_A and 2theta_B lie in 0, pi , so r_A > 0 and r_B > 0 . Let theta_A = alpha ; then theta_B = alpha + frac{pi}{3} , where 0 < alpha < frac{pi}{6} . Substituting r_A and r_B : r_A = 4sin2alpha, quad r_B = 4sinleft2alpha + frac{2pi}{3}right text{Area} = frac{sqrt{3}}{4} cdot 16sin2alphasinleft2alpha + frac{2pi}{3}right = 4sqrt{3}sin2alphasinleft2alpha + frac{2pi}{3}right Using the product-to-sum identity sinxsiny = frac{1}{2}[cosx - y - cosx + y] : sin2alphasinleft2alpha + frac{2pi}{3}right = frac{1}{2}left[cosleft-frac{2pi}{3}right - cosleft4alpha + frac{2pi}{3}rightright] = frac{1}{2}left[-frac{1}{2} - cosleft4alpha + frac{2pi}{3}rightright] Since 0 < alpha < frac{pi}{6} , the angle 4alpha + frac{2pi}{3} ranges across leftfrac{2pi}{3}, frac{4pi}{3}right . The cosine function reaches its minimum of -1 at 4alpha + frac{2pi}{3} = pi implies 4alpha = frac{pi}{3} implies alpha = frac{pi}{12} , which lies inside the interval. At this value, the product attains its maximum: frac{1}{2}left[-frac{1}{2} - -1right] = frac{1}{4} Thus, the maximum possible area of triangle OAB is: text{Area}_{max} = 4sqrt{3} cdot frac{1}{4} = sqrt{3} A common error is to assume both r_A and r_B can independently reach the maximum curve radius of 4 , which would incorrectly yield frac{sqrt{3}}{4} cdot 4 cdot 4 = 4sqrt{3} . However, because theta_B - theta_A = frac{pi}{3} , the radii are coupled and cannot both equal 4 simultaneously.

(c) ClipSAT. By Mr. Mohamed Adaabah +96597685887

Unit 5.01: Trigonometry, Identities & Polar Coordinates

Q4. A point in the polar coordinate system has coordinates r, theta = left6, frac{5pi}{6}right. Which of the following represents the Cartesian coordinates x, y of this point?

[x] A: -3sqrt{3}, 3

[ ] B: -3, 3sqrt{3}

[ ] C: 3sqrt{3}, -3

[ ] D: -3sqrt{3}, -3

Solution: To convert polar coordinates r, theta to Cartesian coordinates x, y, use the conversion formulas: x = r costheta quad text{and} quad y = r sintheta Given r = 6 and theta = frac{5pi}{6}: x = 6 cosleftfrac{5pi}{6}right = 6 left-frac{sqrt{3}}{2}right = -3sqrt{3} y = 6 sinleftfrac{5pi}{6}right = 6 leftfrac{1}{2}right = 3 Thus, the Cartesian coordinates of the point are -3sqrt{3}, 3. Common errors include swapping the cosine and sine components resulting in -3, 3sqrt{3} or making sign errors with Quadrant II trigonometric values.

Q5. In triangle ABC, the lengths of sides AB and BC are 5 and 8, respectively, and the measure of angle B is 60^circ. What is the length of the altitude from vertex B to side AC?

[x] A: frac{20sqrt{3}}{7}

[ ] B: frac{40sqrt{3}}{7}

[ ] C: frac{10sqrt{3}}{7}

[ ] D: frac{20}{7}

Solution: To find the altitude from vertex B to side AC, we first determine the length of side AC using the Law of Cosines: AC^2 = AB^2 + BC^2 - 2ABBCcosangle B Substitute the given values: AC^2 = 5^2 + 8^2 - 258cos60^circ = 25 + 64 - 80leftfrac{1}{2}right = 89 - 40 = 49 Taking the positive square root gives AC = 7. Next, calculate the area of triangle ABC using the sine formula for triangle area: text{Area} = frac{1}{2} cdot AB cdot BC cdot sinangle B = frac{1}{2} cdot 5 cdot 8 cdot sin60^circ = 20 cdot frac{sqrt{3}}{2} = 10sqrt{3} Finally, relate the area to the base AC and the altitude h from B: text{Area} = frac{1}{2} cdot AC cdot h implies 10sqrt{3} = frac{1}{2} cdot 7 cdot h implies h = frac{20sqrt{3}}{7} Common errors: - frac{40sqrt{3}}{7} results from omitting the factor of frac{1}{2} in the area formula. - frac{10sqrt{3}}{7} results from computing frac{text{Area}}{AC} instead of frac{2cdottext{Area}}{AC}. - frac{20}{7} results from mistakenly using cos60^circ = frac{1}{2} instead of sin60^circ = frac{sqrt{3}}{2} when finding the area.

Q6. In the polar coordinate system, point A has polar coordinates r_A, theta_A lying on the curve r = 6sintheta, and point B has polar coordinates r_B, theta_B lying on the curve r = 8costheta, with r_A > 0 and r_B > 0. If theta_A - theta_B = frac{pi}{6}, what is the maximum possible area of triangle OAB, where O is the pole origin?

[ ] A: 6

[ ] B: 6sqrt{3}

[x] C: 9

[ ] D: 12

Solution: In polar coordinates, the area of a triangle with one vertex at the pole O0, 0 and the other two vertices at Ar_A, theta_A and Br_B, theta_B is given by: text{Area}triangle OAB = frac{1}{2} r_A r_B sin|theta_A - theta_B| We are given theta_A - theta_B = frac{pi}{6}, so sin|theta_A - theta_B| = sinleftfrac{pi}{6}right = frac{1}{2}. Let theta_B = theta. Then theta_A = theta + frac{pi}{6}. Since point A lies on r = 6sintheta and point B lies on r = 8costheta, we have: r_A = 6sinleftttheta + frac{pi}{6}right, quad r_B = 8costheta Substituting these expressions into the area formula: text{Area} = frac{1}{2} left[6sinleftttheta + frac{pi}{6}rightright]8costhetaleftfrac{1}{2}right = 12sinleftttheta + frac{pi}{6}rightcostheta Using the product-to-sum trigonometric identity sin u cos v = frac{1}{2}[sinu + v + sinu - v] with u = theta + frac{pi}{6} and v = theta: sinleftttheta + frac{pi}{6}rightcostheta = frac{1}{2}left[sinleft2theta + frac{pi}{6}right + sinleftfrac{pi}{6}rightright] = frac{1}{2}left[sinleft2theta + frac{pi}{6}right + frac{1}{2}right] Thus, the area simplifies to: text{Area} = 12 cdot frac{1}{2}left[sinleft2theta + frac{pi}{6}right + frac{1}{2}right] = 6sinleft2theta + frac{pi}{6}right + 3 The maximum value of sinleft2theta + frac{pi}{6}right is 1, which occurs when 2theta + frac{pi}{6} = frac{pi}{2} implies

Unit 5.01: Trigonometry, Identities & Polar Coordinates

Q7. A point in the polar coordinate plane has coordinates r, theta = left4, frac{5pi}{6}right . Which of the following gives the rectangular coordinates x, y of this point?

[x] A: -2sqrt{3}, 2

[ ] B: -2, 2sqrt{3}

[ ] C: 2, -2sqrt{3}

[ ] D: 2sqrt{3}, -2

Solution: To convert polar coordinates r, theta to rectangular coordinates x, y , use the conversion formulas: x = r cos theta quad text{and} quad y = r sin theta Given r = 4 and theta = frac{5pi}{6} : x = 4 cosleftfrac{5pi}{6}right = 4 left-frac{sqrt{3}}{2}right = -2sqrt{3} y = 4 sinleftfrac{5pi}{6}right = 4 leftfrac{1}{2}right = 2 Thus, the rectangular coordinates are -2sqrt{3}, 2 . - Choice B is incorrect because it mistakenly swaps the values of cosleftfrac{5pi}{6}right and sinleftfrac{5pi}{6}right . - Choice C is incorrect because it swaps the x - and y -coordinates. - Choice D is incorrect because it reverses the signs of both coordinates.

Q8. In triangle ABC , the side lengths are AB = 5 and BC = 4 , and angle B is an acute angle. If cos2B = -frac{31}{32} , what is the length of side AC ?

[x] A: 6

[ ] B: sqrt{46}

[ ] C: sqrt{41}

[ ] D: sqrt{31}

Solution: We solve the problem in two steps by first determining cosB and then applying the Law of Cosines to find AC . **Step 1: Find cosB ** Using the double-angle identity for cosine: cos2B = 2cos^2B - 1 Substitute the given value cos2B = -frac{31}{32} : -frac{31}{32} = 2cos^2B - 1 2cos^2B = 1 - frac{31}{32} = frac{1}{32} cos^2B = frac{1}{64} Since angle B is acute 0^circ < B < 90^circ , cosB > 0 : cosB = frac{1}{8} **Step 2: Apply the Law of Cosines** In triangle ABC , the side opposite angle B is AC : AC^2 = AB^2 + BC^2 - 2ABBCcosB Substitute AB = 5 , BC = 4 , and cosB = frac{1}{8} : AC^2 = 5^2 + 4^2 - 254leftfrac{1}{8}right AC^2 = 25 + 16 - frac{40}{8} = 41 - 5 = 36 Taking the principal square root gives: AC = 6 **Common Traps:** - sqrt{46} arises from using an incorrect sign in the Law of Cosines 41 + 5 = 46 or erroneously setting cosB = -frac{1}{8} . - sqrt{41} comes from assuming triangle ABC is a right triangle and applying the Pythagorean theorem directly sqrt{5^2 + 4^2} . - sqrt{31} results from an arithmetic error such as computing cosB = frac{1}{4} instead of frac{1}{8} .